

2.6 LOWEST COMMON MULTIPLE (LCM)

Determine common multiples of two or more 2-digit numbers

The multiples of any two numbers which are common in both the numbers are called common multiples.

Example 1:

Let's start with the numbers 4 and 6:

Solution:

Multiples of 4 are: 4, 8, **12**, 16, 20, **24**, 28, 32, **36**, ...

Multiples of 6 are: 6, **12**, 18, **24**, 30, **36**, 42, 48, 54, ...

The common multiples of 4 and 6 are **12**, **24**, **36**, ...

There is no end to common multiples of any two numbers because multiples are not limited!!!

Example 2:

Determine first 4 common multiples of 8, 12 and 24

Solution:

Multiples of 8 are: 8, 16, **24**, 32, 40, **48**, 56, 64, **72**, 80, 88, **96**, ...

Multiples of 12 are: 12, **24**, 36, **48**, 60, **72**, 84, **96**, 108, ...

Multiples of 24 are: **24**, **48**, **72**, **96**, ...

We can see that the first four common multiples of 8, 12

and 24 are **24**, **48**, **72** and **96**

Find Lowest Common Multiple by

(i) Common multiples (ii) Prime factorization

In order to find the lowest common multiple (LCM) of two numbers, consider the above examples:

In example 1, out of the common multiples of 4 and 6, the lowest is **12**, so **12** is the Lowest Common Multiple of 4 and 6.

Will the highest common multiple exist????

Similarly in example 2, out of the common multiples of 8 and 12, the lowest is **24**, so **24** is the Lowest Common Multiple of 8 and 12.

This method of finding LCM is called the **common multiples method**. The other method is **Prime factorization method**.

Example 1: Find the Lowest Common Multiple of 24, 36 and 48 by prime factorization method.

Solution:

2	24
2	12
2	6
3	3
	1

2	36
2	18
3	9
3	3
	1

2	48
2	24
2	12
2	6
3	3
	1

$$\begin{aligned}
 24 &= 2 \times 2 \times 2 \times 3 \\
 36 &= 2 \times 2 \times 3 \times 3 \\
 48 &= 2 \times 2 \times 2 \times 2 \times 3
 \end{aligned}$$

LCM must be divisible by all the numbers given

LCM = Product of common factors $\times$ all uncommon factors.
 $= 2 \times 2 \times 2 \times 3 \times 3 \times 2 = 144$

Example 2: Find the LCM of 12, 18 and 20 using the Prime Factorization method.



$$12 = 2 \times 2 \times 3$$

$$18 = 2 \times 3 \times 3$$

$$20 = 2 \times 2 \times 5$$

$$\text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 1080$$

EXERCISE 2.7

1. Find the LCM of the following numbers by common multiples method:

- (i) 25 and 15 (ii) 12 and 14 (iii) 10 and 20
(iv) 6 and 9 (v) 7 and 5

2. Find the LCM of the following numbers by prime factorization method:

- (i) 42 and 18 (ii) 15 and 36
(iii) 12 and 45 (iv) 35 and 15
(v) 20 and 48 (vi) 27 and 24
(vii) 45 and 63 (viii) 30 and 18
(ix) 11, 22 and 33 (x) 24, 30 and 36

Solve real life problems involving LCM**Example 1:**

Today, both the soccer team and the basketball team had games. The soccer team plays every 3 days and the basketball team plays every 5 days. When will both teams have games on the same day again?

Solution:

Lowest common multiple of 3 and 5 is 15
After 15 days they both will have games on the same day.

Example 2:

A manager at a restaurant can buy buns in packages of 8, Burger in the packages of 10 and shami kababs in packages of 6. What is the least number of packages of each product he should buy to have an equal number of buns and kababs?

Solution:

Lowest common multiple of 8, 10 and 6 is 120
He should buy 120 packages of each product to have equal number of buns and kababs.

Example 3. There are three bells ring at an interval of 7, 10 and 14 minutes respectively. They begin by ringing to gather. Find out how long will they ring to gather again?

Solution: Interval of three bells is 7, 10 and 14 minutes

LCM of (7, 10, 14) minutes

LCM = 70 minutes

Therefore, They will ring to gather after 70 minutes.

2	7, 10, 14
5	7, 5, 7
7	7, 1, 7
	1, 1, 1

$$\text{L.C.M} = 2 \times 5 \times 7 = 70$$

EXERCISE 2.8

1. Ali saves Rs. 9 every day while Maria saves Rs.12. What is the least number of days it will take each person to save the same amount of money?
2. A child smiles at another child every 3 seconds while the other child smiles back every 6 seconds. When will both children smiles at each other at the same time?
3. Boxes that are 12 inches tall are being piled next to boxes that are 10 inches tall. What is the least height in feet at which the two piles will be the same height?
4. Two men running a marathon took a sip of water at the same time 72 minutes after they started the race. If the first man took a sip of water every 9 minutes, how often did the other man take a sip of water?
5. A radio station plays "yesterday" by the Beatles once every 2 days. Another station plays the same song once every 3 days. How many times in 30 days will both radio stations play the same song on the same day?
6. Ali runs every 12 days, Anees every 8 days and Ameen every 6 days. Ali Anees and Ameen boht ran today. How many days will it be until run together again?
7. Ali visits the garden every 15 days and Azhar every 20 days. Ali and Azhar both meet today. After how many days they will meet in garden again?

REVIEW EXERCISE – 2**1. Choose the correct answer:**

(i) Which of the following number is divisible by 2?

- (a) 567 (b) 484 (c) 257 (d) 193

(ii) Which of the following number is divisible by 3?

- (a) 143 (b) 483 (c) 367 (d) 941

(iii) Which of the following number is divisible by 2 and 10 both?

- (a) 4579 (b) 3921 (c) 4050 (d) 2108

(iv) What is the HCF of 39 and 52?

- (a) 39 (b) 26 (c) 13 (d) 1

(v) What is the LCM of 48 and 66?

- (a) 528 (b) 185 (c) 246 (d) 114

2. Find the HCF and LCM of 48, 69 and 36?

3. What is the HCF and LCM of 37 and 41? Explain how you got your answer?

4. Mrs. Sayem wants to distribute 45 patties, 60 samosas and 75 juices equally. Find the greatest number of students among which these materials can be distributed equally.

5. Rida is arranging her birthday party in cafeteria. She wants to share 15 sandwiches or 30 slices of pizza to be equally distributed in each friend. Find the number of friends could be invited at the party?

3.1 FRACTIONS

Define a fraction

Fraction is the part of a whole which is equally divided.



$$\frac{3}{4}$$

← Numerator

← Denominator

Number of parts taken

Total number of parts

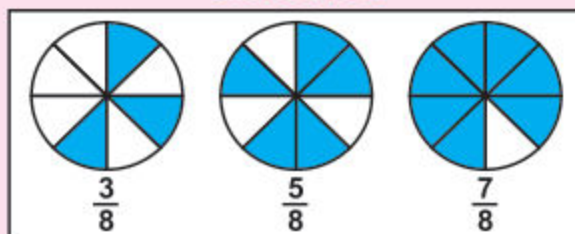
Examples: Coloured parts of a whole are shown fraction.

One Whole 	Five - Sixths
One - Half 	Six - Sevenths
Two - Thirds 	Seven - Eighths
Three - Fourths 	Eight - Ninths
Four - Fifth 	Nine - Tenths

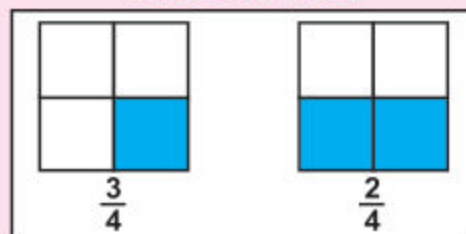
Recognize like and unlike fractions

Look at the figures below:

Coloured



Uncoloured



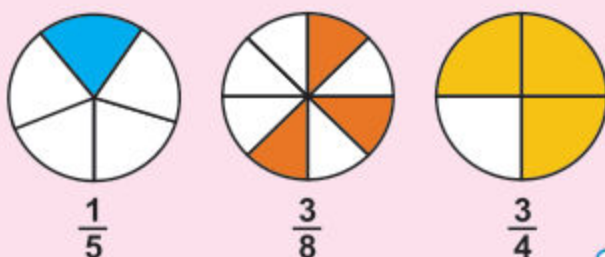
What do you observe?

One figure has been divided into equal parts.

The denominator in each case is the same.

They are called **Like fractions**.Fractions having different denominators are called **Unlike fractions**.

A whole is divided into same number of parts.



A whole is divided into different number of parts.

One-fifth, three-eighths and three-fourths are unlike fractions

Example: Separate the pairs of like and unlike fractions.

(i) $\frac{3}{4}, \frac{5}{7}$ (ii) $\frac{1}{5}, \frac{2}{5}$

Solution:

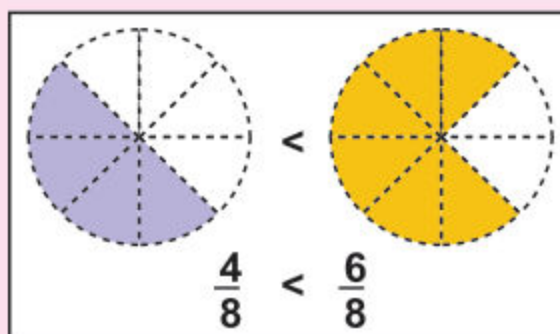
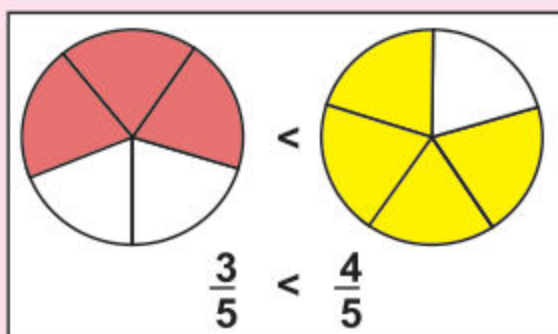
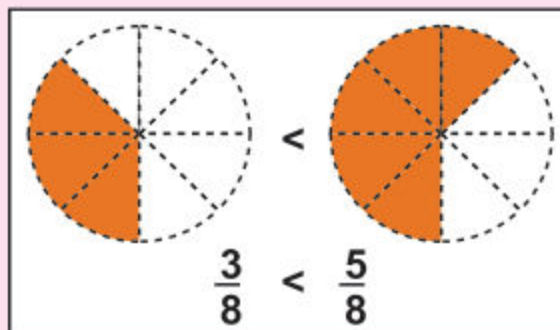
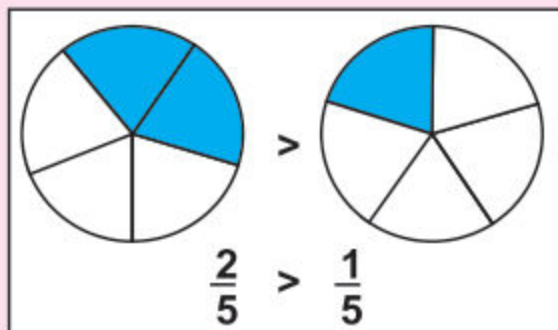
Like fraction pair: $\frac{1}{5}, \frac{2}{5}$

Unlike fraction pair: $\frac{3}{4}, \frac{5}{7}$

Compare two unlike fractions by converting them to equivalent fractions with the same denominator.

In the previous class you have learnt how to compare like fractions. Let us recall how you did that?

Compare fractions represented by coloured portion.



In case of like fractions, just look at the numerator, fraction with greater numerator represents greater value!

Examples:

$$\frac{8}{9} > \frac{5}{9} \quad , \quad \frac{7}{11} > \frac{2}{11} \quad , \quad \frac{1}{16} < \frac{5}{16} \quad , \quad \frac{2}{7} < \frac{5}{7}$$

We can compare unlike fractions as well.

Example:

How would you know whether $\frac{3}{4}$ is less than or greater than $\frac{4}{7}$?

Solution:

Convert these fractions into equivalent fractions having same denominator.

Equivalent fractions are obtained by multiplying the numerator and denominator by same number. It does not change the value of fraction !!!

LCM of 4 and 7 is 28

Now we make denominators same

$$\frac{3}{4} = \frac{3 \times 7}{4 \times 7} = \frac{21}{28}$$

$$\frac{4}{7} = \frac{4 \times 4}{7 \times 4} = \frac{16}{28}$$

Now they are like fractions, so just compare the numerators.

Here,

$$\frac{21}{28} > \frac{16}{28}$$

So, $\frac{3}{4} > \frac{4}{7}$

REMEMBER!

- Take the LCM of denominators
- Multiply the numerator and denominator by the same number to convert the denominator into LCM

Arrange fractions in ascending and descending order

Example 1:

Arrange the following sets of fractions in ascending and descending order.

$$\frac{6}{7}, \frac{2}{7}, \frac{1}{7} \text{ and } \frac{3}{7}$$

Solution: $\frac{6}{7}, \frac{2}{7}, \frac{1}{7}, \frac{3}{7}$

Since they are like fractions, compare the numerators

Ascending order:

$$\frac{1}{7}, \frac{2}{7}, \frac{3}{7}, \frac{6}{7}$$

Descending order:

$$\frac{6}{7}, \frac{3}{7}, \frac{2}{7}, \frac{1}{7}$$

Example 2: Compare $\frac{5}{9}$ and $\frac{4}{5}$

Solution:

LCM of 5 and 9 is 45

So, $\frac{5}{9} = \frac{5 \times 5}{9 \times 5} = \frac{25}{45}$ We make the denominator of both fractions 45

And $\frac{4}{5} = \frac{4 \times 9}{5 \times 9} = \frac{36}{45}$ So, we compare numerators 25 and 36

Because, $25 < 36$

So, $\frac{5}{9} < \frac{4}{5}$

Example 3: Write the following fractions in both ascending and descending order.

$\frac{2}{9}$, $\frac{5}{6}$, $\frac{7}{12}$ and $\frac{1}{3}$

Solution:

Since they are unlike fractions, we will first have to convert them into equivalent fractions.

STEP 1: Take the LCM of denominators
(LCM of 9, 6, 12 and 3 is 36)

STEP 2: Multiply numerator and denominator by the same number to get the LCM in the denominator.

$$\frac{2 \times 4}{9 \times 4} = \frac{8}{36}, \frac{5 \times 6}{6 \times 6} = \frac{30}{36}, \frac{7 \times 3}{12 \times 3} = \frac{21}{36}, \frac{1 \times 12}{3 \times 12} = \frac{12}{36}$$

$$\text{Or } \frac{8}{36}, \frac{30}{36}, \frac{21}{36}, \frac{12}{36}$$

Now it is easier to arrange by looking at the numerators,

Ascending order:

$$\frac{8}{36}, \frac{12}{36}, \frac{21}{36}, \frac{30}{36}$$

Or $\frac{2}{9}, \frac{1}{3}, \frac{7}{12}, \frac{5}{6}$

If the denominators are same, greater the numerator, larger the value of the fraction.

Descending order:

$$\frac{5}{6}, \frac{7}{12}, \frac{1}{3}, \frac{2}{9}$$

Simplify fractions to the lowest form

Fractions can be reduced to lowest form by dividing numerator and denominator by the highest common factor.

Example 1: Simplify: $\frac{30}{45}$

Solution: $\frac{30}{45} \rightarrow$ HCF of 30 and 45 is 15

$$\frac{30 \div 15}{45 \div 15} = \frac{2}{3}$$

So, $\frac{30}{45} = \frac{10}{15} = \frac{2}{3}$, where $\frac{2}{3}$ is the lowest form.

Teacher's Note

Teacher should teach the students about the ascending and descending orders of fractions by using LCM.

EXERCISE 3.1

1. From the following pairs of fractions, identify pair of like or unlike fractions.

(i) $\frac{2}{7}$ and $\frac{6}{7}$ (ii) $\frac{2}{9}$ and $\frac{2}{15}$ (iii) $\frac{8}{17}$ and $\frac{7}{17}$

(iv) $\frac{3}{10}$ and $\frac{6}{11}$ (v) $\frac{7}{11}$ and $\frac{7}{10}$ (vi) $\frac{1}{4}$ and $\frac{3}{4}$

(vii) $\frac{4}{13}$ and $\frac{13}{15}$ (viii) $\frac{11}{19}$ and $\frac{16}{19}$ (ix) $\frac{77}{27}$ and $\frac{4}{27}$

2. Fill in the blank boxes with “<” or “>” or “=” by first converting the fractions into like fractions.

(i) $\frac{3}{8}$ $\frac{3}{8}$ (ii) $\frac{5}{6}$ $\frac{4}{9}$ (iii) $\frac{7}{12}$ $\frac{1}{4}$

(iv) $\frac{2}{7}$ $\frac{13}{14}$ (v) $\frac{6}{15}$ $\frac{3}{10}$ (vi) $\frac{9}{11}$ $\frac{16}{22}$

3. Arrange the following fractions in ascending order.

(i) $\frac{6}{7}$, $\frac{9}{7}$, $\frac{5}{7}$

(ii) $\frac{5}{6}$, $\frac{2}{3}$, $\frac{5}{9}$

(iii) $\frac{2}{5}$, $\frac{1}{3}$, $\frac{4}{15}$

(iv) $\frac{5}{12}$, $\frac{4}{8}$, $\frac{3}{4}$, $\frac{1}{6}$

(v) $\frac{9}{8}$, $\frac{7}{2}$, $\frac{13}{6}$, $\frac{5}{4}$

(vi) $\frac{5}{12}$, $\frac{7}{6}$, $\frac{7}{4}$, $\frac{5}{2}$

4. Simplify the following fractions to their lowest form.

(i) $\frac{24}{32}$

(ii) $\frac{18}{27}$

(iii) $\frac{30}{50}$

(iv) $\frac{14}{42}$

(v) $\frac{33}{66}$

3.2 TYPES OF FRACTION

Unit Fractions: numerator is 1

Example:



$$\frac{1}{2}$$



$$\frac{1}{5}$$



$$\frac{1}{4}$$

numerator is
equal to one.

Non-unit Fractions: numerator is more than 1

Example:



$$\frac{3}{4}$$



$$\frac{3}{5}$$

Proper Fractions: numerator is less than denominator

Example:

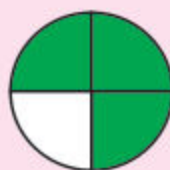
Proper Fractions

Numerator is less than
the Denominator.

$$\frac{1}{5}$$



$$\frac{3}{8}$$



$$\frac{3}{4}$$



Improper Fractions: numerator is equal or greater than denominator

Example:

Improper Fractions

Numerator is greater
than or equal to the
Denominator.

$$\frac{3}{3} + \frac{1}{3} = \frac{4}{3}$$

$$\frac{4}{4} + \frac{3}{4} = \frac{7}{4}$$

They are
greater than
or equal to 1

Mixed Numbers: a whole number and a proper fraction together

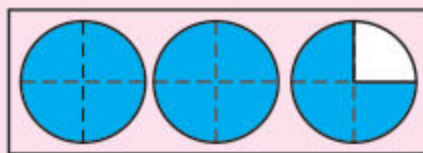
Example:

Mixed Fractions

A whole and a proper fraction



$$1\frac{1}{2}$$



$$2\frac{3}{4}$$

They are similar to improper fractions



$$1\frac{1}{3}$$

Convert improper fraction to mixed fraction and vice versa

Example 1:

Convert $5\frac{1}{2}$ into an improper fraction

5 wholes = 10 halves

10 halves + 1 half = 11 halves

$10 + 1 = 11$

$$5\frac{1}{2} = \frac{11}{2}$$

Example 2:

Convert $\frac{13}{4}$ into an mixed number

Four quarters = one whole

13 quarters = 3 wholes and a quarter remaining

($13 \div 4 = 3$ remainder 1)

$$\frac{13}{4} = 3\frac{1}{4}$$

EXERCISE 3.2

1. Identify which of the following are unit, proper, improper or mixed fractions?

(i) $\frac{1}{3}$

(ii) $\frac{1}{4}$

(iii) $\frac{19}{4}$

(iv) $\frac{1}{7}$

(v) $\frac{5}{9}$

(vi) $2\frac{1}{8}$

(vii) $\frac{4}{3}$

(viii)



(ix)



(x)



2. Convert the following improper fractions into mixed fractions.

(i) $\frac{43}{7}$

(ii) $\frac{29}{4}$

(iii) $\frac{74}{9}$

(iv) $\frac{14}{6}$

(v) $\frac{28}{5}$

(vi) $\frac{67}{3}$

(vii) $\frac{85}{11}$

(viii) $\frac{86}{5}$

3. Express the following mixed fractions as improper fractions.

(i) $4\frac{3}{10}$

(ii) $5\frac{2}{3}$

(iii) $6\frac{1}{2}$

(iv) $3\frac{2}{7}$

(v) $8\frac{1}{4}$

(vi) $1\frac{11}{13}$

(vii) $7\frac{7}{8}$

(viii) $2\frac{6}{7}$

3.3 ADDITION AND SUBTRACTION OF FRACTIONS

In the previous class, you have learnt how to add and subtract like fractions, let us recall that.

Example 1: Add: $\frac{2}{7} + \frac{3}{7}$

Solution:

$$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$$

Example 2: Solve: $\frac{7}{11} - \frac{4}{11}$

Solution:

$$\frac{7}{11} - \frac{4}{11} = \frac{3}{11}$$



Activity

Solve.

$$\frac{3}{7} + \frac{5}{7} = \boxed{}$$

$$\frac{4}{5} - \frac{2}{5} = \boxed{}$$

$$\frac{2}{9} + \frac{3}{9} = \boxed{}$$

$$\frac{9}{15} - \frac{8}{15} = \boxed{}$$

$$\frac{5}{7} + \frac{1}{7} = \boxed{}$$

$$\frac{7}{15} - \frac{4}{15} = \boxed{}$$

$$\frac{6}{10} + \frac{3}{10} = \boxed{}$$

$$\frac{7}{12} - \frac{6}{12} = \boxed{}$$

$$\frac{5}{12} + \frac{2}{12} = \boxed{}$$

$$\frac{8}{14} - \frac{5}{14} = \boxed{}$$

$$\frac{7}{11} + \frac{2}{11} = \boxed{}$$

$$\frac{13}{17} - \frac{4}{17} = \boxed{}$$

Add fractions with unlike denominators

Example 1: Add $\frac{2}{3} + \frac{5}{6}$

Solution: We will first have to convert these unlike fractions into like fractions by taking the LCM of denominators.

Lowest common multiple of 3 and 6 is 6

$$\frac{2}{3} = \frac{4}{6} \text{ (using equivalent fractions)}$$

$$\text{So, } \frac{2}{3} + \frac{5}{6}$$

$$= \frac{4}{6} + \frac{5}{6} = \frac{4+5}{6} \text{ (add the numerators)}$$

$$= \frac{9}{6} = \frac{3}{2} = 1\frac{1}{2} \text{ (always keep your answer in the lowest form)}$$

Teacher's Note

Teacher should put emphasis on the use of LCM in addition and subtraction of fractions with unlike denominators.

Unit 3 FRACTIONS (Addition and Subtraction of Fractions)

Example 2: Add $1\frac{3}{5} + 2\frac{1}{10}$

Solution:

$$1\frac{3}{5} + 2\frac{1}{10} = \frac{8}{5} + \frac{21}{10}$$

Convert into improper fractions

$$= \frac{8 \times 2}{5 \times 2} + \frac{21 \times 1}{10 \times 1}$$

Make them like fractions (LCM = 10)

$$= \frac{16}{10} + \frac{21}{10} = \frac{16 + 21}{10}$$

Do the addition of numerators

$$= \frac{37}{10}$$

$$= 3\frac{7}{10}$$

Always keep your answer in lowest form

$$\begin{array}{r} 10) 37 (3 \\ - 30 \\ \hline 7 \end{array}$$

Verify commutative property of addition of fraction with same denominator

Example 1: Verify commutative property of addition for $\frac{5}{7}$ and $\frac{6}{7}$.

Commutative property of addition

$$\frac{5}{7} + \frac{6}{7} = \frac{6}{7} + \frac{5}{7}$$

$$\text{L.H.S} = \frac{5}{7} + \frac{6}{7}$$

$$= \frac{5+6}{7} = \frac{11}{7}$$

Now

$$\text{R.H.S} = \frac{6}{7} + \frac{5}{7}$$

$$= \frac{6+5}{7} = \frac{11}{7}$$

$$\text{L.H.S} = \text{R.H.S} \quad \text{verified}$$

Verify associative property of addition of fraction with same denominator

Example 2: Verify associative property of addition for

$$\frac{1}{2}, \frac{5}{2} \text{ and } \frac{9}{2}$$

Unit 3 FRACTIONS (Addition and Subtraction of Fractions)

Associative property of addition

$$\begin{aligned} & \left(\frac{1}{2} + \frac{5}{2} \right) + \frac{9}{2} = \frac{1}{2} + \left(\frac{5}{2} + \frac{9}{2} \right) \\ \text{L.H.S} &= \left(\frac{1}{2} + \frac{5}{2} \right) + \frac{9}{2} \\ &= \left(\frac{1+5}{2} \right) + \frac{9}{2} \\ &= \frac{6}{2} + \frac{9}{2} \\ \text{L.H.S} &= \frac{6+9}{2} = \frac{15}{2} \\ \text{R.H.S} &= \frac{1}{2} + \left(\frac{5}{2} + \frac{9}{2} \right) \\ &= \frac{1}{2} + \frac{14}{2} \\ \text{R.H.S} &= \frac{1+14}{2} = \frac{15}{2} = \text{L.H.S verified} \end{aligned}$$

Subtract fractions with unlike denominators

We know that while subtraction of fractions with same denominator (like fractions) only numerators are subtracted and denominator of the difference remains the same, as of the given fraction.

Example 1: Subtract $\frac{5}{8} - \frac{1}{4}$

$$\begin{aligned} & \frac{5}{8} - \frac{1}{4} \quad (\text{Making like denominators}) \\ &= \frac{5}{8} - \frac{2}{8} = \frac{5-2}{8} \quad (\text{Take LCM of 8 and 4}) \\ &= \frac{3}{8} \quad (\text{subtract nominators}) \end{aligned}$$

Example 2: Subtract $3\frac{3}{4} - 1\frac{1}{6}$

Convert the mixed fractions into improper fraction

Unit 3 FRACTIONS (Addition and Subtraction of Fractions)

$$3 \frac{3}{4} = \frac{15}{4} \text{ and } 1 \frac{1}{6} = \frac{7}{6}$$

$$3 \frac{3}{4} - 1 \frac{1}{6} = \frac{15}{4} - \frac{7}{6} \quad (\text{LCM of 4 and 6 is 12})$$

$$= \frac{15}{4} = \frac{45}{12} \text{ and } \frac{7}{6} = \frac{14}{12} \quad (\text{like fraction})$$

$$3 \frac{3}{4} - 1 \frac{1}{6} = \frac{45}{12} - \frac{14}{12} = \frac{31}{12}$$

$$= 2 \frac{7}{12}$$

EXERCISE 3.3

1. Add the following fractions.

$$(i) \frac{2}{3} + \frac{11}{9}$$

$$(ii) \frac{7}{6} + \frac{1}{2}$$

$$(iii) \frac{8}{10} + \frac{1}{2} + \frac{2}{4}$$

$$(iv) \frac{3}{4} + \frac{1}{2} + \frac{4}{6}$$

$$(v) 3 \frac{1}{2} + 5 \frac{1}{3}$$

$$(vi) 5 \frac{1}{3} + 2 \frac{3}{4}$$

$$(vii) \frac{2}{3} + 1 \frac{1}{5} + \frac{3}{2}$$

$$(viii) 2 \frac{1}{7} + \frac{2}{5} + 1 \frac{1}{7}$$

2. Subtract the following fractions.

$$(i) \frac{5}{6} - \frac{1}{2} \quad (ii) \frac{7}{8} - \frac{3}{4} \quad (iii) \frac{4}{5} - \frac{1}{3}$$

$$(iv) 3 \frac{3}{10} - 1 \frac{1}{4} \quad (v) 9 \frac{1}{2} - 3 \frac{1}{5} \quad (vi) \frac{4}{5} - \frac{1}{10} - \frac{2}{15}$$

3. Apply commutative and associative property of addition to fill in the blanks.

$$(i) \frac{3}{5} + \frac{7}{4} = \dots\dots\dots + \frac{3}{5} \quad (ii) \frac{2}{9} + \dots\dots\dots = \frac{4}{7} + \dots\dots\dots$$

$$(iii) \frac{1}{3} + \left(\frac{1}{5} + \frac{1}{4} \right) = \left(\frac{1}{3} + \dots\dots\dots \right) + \frac{1}{4}$$

$$(iv) \frac{3}{10} + \left(\dots\dots\dots + \frac{4}{5} \right) = \left(\dots\dots\dots + \frac{2}{7} \right) + \frac{4}{5}$$

3.4 MULTIPLICATION OF FRACTIONS

Multiply fractions with whole numbers

Consider the diagram



Shaded part in each figure represents $\frac{1}{8}$

$\frac{3}{8}$ Parts are shaded altogether

Or $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$

Or $3 \times \frac{1}{8} = \frac{3}{8}$

In multiplying a whole number with a fraction, multiply the whole number with the numerator

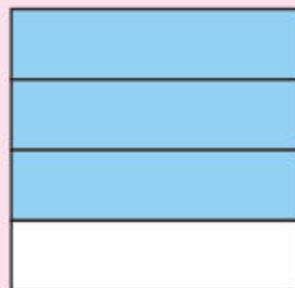
Example 1	Steps Followed	Example 2
$5 \times \frac{3}{4}$ $= \frac{5 \times 3}{4}$ $= \frac{15}{4}$ $= 3\frac{3}{4}$	▪ Multiply the numerator with the whole number. ▪ Reduce it, if possible. ▪ Convert into mixed number (if improper). Write your answer in simplest form.	$7 \times \frac{3}{14}$ $= \frac{21}{14}$ $= \frac{\overset{3}{\cancel{21}}}{\underset{2}{\cancel{14}}}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$

Multiply two or more fractions (proper fractions, improper fractions and mixed numbers)

This figure represents $\frac{3}{4}$.

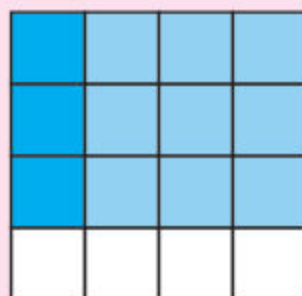
In order to find $\frac{1}{4}$ of $\frac{3}{4}$,

We will further divide $\frac{3}{4}$ into four equal parts and shade one out of four parts.



The figure below represents $\frac{1}{4}$ of $\frac{3}{4}$ which is $\frac{3}{16}$ (the double shaded region).

This was the pictorial representation of $\frac{1}{4}$ of $\frac{3}{4}$



Mathematically, $\frac{1}{4}$ of $\frac{3}{4} = \frac{1}{4} \times \frac{3}{4}$

$$= \frac{1 \times 3}{4 \times 4} \longrightarrow \text{Multiply the numerator and denominator}$$

$$= \frac{3}{16} \longrightarrow \text{Write the product in lowest term}$$

Example : Solve: $\frac{2}{9} \times \frac{3}{5} \times \frac{1}{2}$

Solution: $\frac{2}{9} \times \frac{3}{5} \times \frac{1}{2}$

$$= \frac{2 \times 3 \times 1}{9 \times 5 \times 2} \longrightarrow \text{Multiply the numbers at numerators and denominators}$$

$$= \frac{6}{90} \div \frac{3}{3}$$

$$= \frac{6 \div 3}{90 \div 3} = \frac{\overset{1}{\cancel{2}}}{\underset{15}{\cancel{30}}} = \frac{1}{15}$$

Teacher's Note

Teacher should develop the concept of multiplication of fractions with the help of geometrical figures.

Multiplying mixed numbers:

Multiplying mixed numbers just requires one extra step i.e.

Must Convert Mixed Number into Improper Fractions First,

Rest of the steps are same as multiplying two or more fractions.

Example 1	Steps Followed	Example 2
$1\frac{7}{3} \times 3\frac{1}{9}$		$4\frac{2}{5} \times 3\frac{7}{11}$
$= \frac{10}{3} \times \frac{28}{9}$	Convert mixed numbers into improper fractions and reduce (if possible)	$= \frac{22}{5} \times \frac{40}{11}$
$= \frac{10 \times 28}{3 \times 9}$	Multiply the numerators and denominators	$= \frac{22 \times 40}{5 \times 11}$
$= \frac{280}{27}$		$= \frac{880}{55}$
$= 10\frac{10}{27}$	Convert the product into mixed number (if improper fraction) Leave your answer in simplest form	$= \frac{176}{11}$
		$= 16$

EXERCISE 3.4**1. Simplify the following fractions.**

(i) $\frac{5}{6}$ of 9 (ii) $\frac{4}{8}$ of 6 (iii) $\frac{3}{14}$ of 45

2. Find the product of the following fractions.

(i) $\frac{1}{2} \times \frac{4}{10}$ (ii) $\frac{6}{9} \times \frac{8}{6}$ (iii) $\frac{15}{21} \times \frac{6}{9} \times \frac{1}{2}$

(iv) $\frac{12}{24} \times \frac{15}{18}$ (v) $\frac{6}{12} \times \frac{21}{24}$ (vi) $\frac{8}{16} \times \frac{4}{8} \times \frac{3}{2}$

(vii) $4\frac{4}{5} \times 4\frac{3}{3}$ (viii) $8\frac{1}{4} \times 3\frac{2}{6}$ (ix) $8\frac{4}{5} \times 3\frac{6}{8}$

3.5 DIVISION OF FRACTIONS

Divide a fraction by a whole number

In order to understand the division of fractions by a whole number, consider the following example.

Example 1: Solve $\frac{1}{2} \div 4$

Solution: $\frac{1}{2} \div 4$ means one half is divided into 4 more parts.

Thus each part will be called $\frac{1}{4}$ of $\frac{1}{2}$.

So, $\frac{1}{4} \times \frac{1}{2}$ or $\frac{1}{2} \times \frac{1}{4}$

$$= \frac{1 \times 1}{2 \times 4}$$

$$= \frac{1}{8}$$

Division by 4 is the same as multiplication by $\frac{1}{4}$ (reciprocal or multiplicative inverse of 4)

Finding Reciprocal of simple fraction means interchanging the numerator and denominator

Example 2: Solve:

Solution:

$$\frac{3}{7} \div 5 = \frac{3}{7} \div \frac{5}{1} \left(5 \text{ is the same as } \frac{5}{1} \right)$$

$$= \frac{3}{7} \times \frac{1}{5} \longrightarrow$$

- Change division into multiplication
- At the same time reciprocate 5

$$= \frac{3 \times 1}{7 \times 5} = \frac{3}{35}$$

Teacher's Note

Teacher should solve some more examples in the class room to strengthen the concept of division of whole number by fraction.

Divide a whole number by a fraction

Example: Solve: $15 \div \frac{2}{3}$

$$\begin{aligned}
 \text{Solution: } 15 \div \frac{2}{3} &= 15 \times \frac{3}{2} \longrightarrow \\
 &= \frac{15 \times 3}{2} \\
 &= \frac{45}{2} = 22\frac{1}{2}
 \end{aligned}$$

Always the number after the division sign is reciprocated

Divide a fraction by another fraction (proper, improper and mixed numbers)

The method remains the same.

Example 1: Solve: $\frac{7}{9} \div \frac{14}{27}$

$$\begin{aligned}
 \text{Solution: } \frac{7}{9} \div \frac{14}{27} &= \frac{\overset{1}{\cancel{7}}}{\underset{1}{\cancel{9}}} \times \frac{\overset{3}{\cancel{27}}}{\underset{2}{\cancel{14}}} \\
 &= \frac{1 \times 3}{1 \times 2} \\
 &= \frac{3}{2} = 1\frac{1}{2}
 \end{aligned}$$

Explanation

- Convert '÷' sign into 'x' sign
- Reciprocates the second fraction
- Multiply the numerators and denominators
- Reduce if possible
- Keep your answer in the simplest form

Example 2: Solve $2\frac{2}{7} \div 1\frac{3}{5}$

$$\begin{aligned}
 \text{Solution: } 2\frac{2}{7} \div 1\frac{3}{5} &= \frac{16}{7} \div \frac{8}{5} \\
 &= \frac{16}{7} \times \frac{5}{8} \\
 &= \frac{\overset{2}{\cancel{16}}}{7} \times \frac{5}{\underset{1}{\cancel{8}}} \\
 &= \frac{10}{7} = 1\frac{3}{7}
 \end{aligned}$$